Week #9 : Cell Membrane Transport — Diffusion, Osmosis & Facilitated Diffusion

Subject/الموضوع: AP Biology Unit Title: Cells / Membrane Transport Grade12: الصف

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can define diffusion and osmosis and compare how particles and water move down concentration gradients. I can explain dynamic equilibrium and the role of aquaporins.	1.1 Flexible display of information (size, color, font, layout).	Week 9 PowerPoint — Lesson 1: Diffusion, Osmosis, Venn Diagram, Aquaporins, AP-style question and exit ticket. Review Teams / AP Biology / General channel
Lesson 2 الحصة الثانية	I can define isotonic, hypotonic and hypertonic solutions. I can predict water movement and explain turgor pressure, lysis and plasmolysis in plant and animal cells.	5.3 Build fluency with graduated scaffolds and models.	Week 9 PowerPoint — Lesson 2: Osmotic Environments, Osmosis Detective Agency, cell diagrams and AP question. Review Teams / AP Biology / General channel
Lesson 3 الحصة الثالثة	I can explain why ions and large polar molecules need facilitated diffusion. I can compare channel proteins and carrier proteins using diagrams, examples and analogies.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Week 9 PowerPoint — Lesson 3: Facilitated Diffusion, Channel vs Carrier Poster, AP-style questions and interactive notes. Review Teams / AP Biology / General channel
Lesson 4 الحصة الرابعة	I can review and integrate membrane structure and transport concepts. I can explain my answers to a peer and identify concepts needing further study.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Week 9 PowerPoint — Membrane Structure and Transport Practice Sheet, interactive notes, collaborative review and reflection. Review Teams / AP Biology / General channel
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	• Biology notebook for Venn diagrams, osmotic cell sketches, channel/carrier poster, practice sheet and reflections. • Microsoft Teams for resources; support with word banks and sentence frames. • Use the practice sheet to identify topics requiring revision.		

الأسبوع# Week #9

Technology Integration دمج التكنولوجيا	• PowerPoint diagrams of diffusion, osmosis and membrane proteins; Microsoft Teams resources. • Optional devices for isotonic/hypertonic/hypotonic research; diagram labeling and group posters.
Cross-curricular links الروابط بين المواد الدراسية	• Chemistry: concentration gradients, polarity, dissolved solutes and equilibrium. • Health: isotonic saline and red blood cell lysis. • UAE agriculture: salt-affected soils and water movement in date palms.
External Assessment Integration دمج التقييمات الخارجية	AP Biology Unit 2: Cells — membrane transport; AP-style questions, osmotic-scenario predictions, membrane-protein comparison and unit practice sheet. Uploaded presentation labels the topic as IB 1.4 Membrane Transport.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: diagram interpretation, compare-and-contrast, prediction and evaluation. • Communication: Venn diagrams, labeled drawings, scientific explanations and poster presentation. • Self-management and social: practice-sheet reflection and collaborative review.
Standards المعايير	• Lesson 1: Membrane Transport — diffusion, osmosis, dynamic equilibrium and aquaporins. • Lesson 2: Osmotic solutions and effects on plant/animal cells. • Lesson 3: Facilitated diffusion through channel and carrier proteins. • Lesson 4: Membrane Structure and Transport — integrated review. • Source PPT identifies IB Standard/Subtopic 1.4 (and 1.3/1.4 for review); AP alignment is thematic.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	Lesson 1: How does water move across membranes? Lesson 2: Why do plant and animal cells respond differently to different osmotic solutions? Lesson 3: If oxygen can diffuse through the membrane, why can't glucose? Lesson 4: How do membrane components and transport processes work together to maintain life?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — Diffusion and Osmosis MUST: Label diffusion and osmosis diagrams. SHOULD: Predict net water movement across membranes. COULD: Explain aquaporins and dynamic equilibrium. Lesson 2 — Osmotic Environments MUST: Identify isotonic, hypotonic and hypertonic solutions. SHOULD: Predict water movement and outcomes in plant/animal cells. COULD: Evaluate claims about lysis versus plasmolysis. Lesson 3 — Facilitated Diffusion MUST: Identify molecules using facilitated diffusion. SHOULD: Compare channels and carriers with examples and diagrams. COULD: Explain protein specificity and relative transport mechanisms.

	Lesson 4 — Review and Practice MUST: Complete membrane transport practice questions. SHOULD: Explain answers and correct misconceptions with a peer. COULD: Integrate membrane structure and transport to justify predictions.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Heritage التراث
	المجال Area: Culture / Innovation / Sustainability
	• Oud incense spreading through air illustrates diffusion in a familiar UAE setting. • Salty soils in date palm agriculture illustrate osmosis and the need for careful irrigation. • Dubai Metro Nol gates provide an analogy for selective membrane-protein pathways. • Urban transport infrastructure provides an analogy for membrane structure and cellular transport.